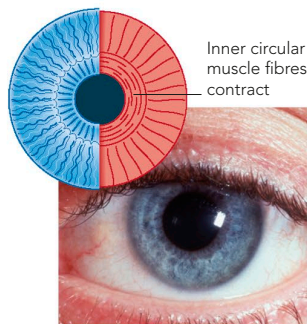


PUPILS

The size of the pupils constantly changes in response to changing levels of light. This is a function of the autonomic nervous system (see pp.106–07). Smooth muscle fibres in the iris are arranged as

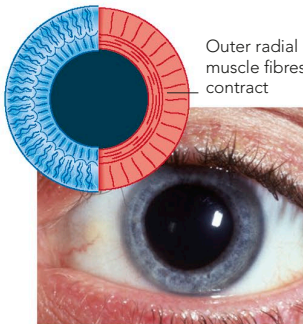
an inner, circular band and an outer, radial band. Sensory receptors in the eyes respond to light and send nerve signals to the brain, which sends messages to one or the other muscle band to adjust pupil size.



Inner circular muscle fibres contract

CONSTRICTED PUPIL

In bright light or to view nearby objects, the pupil constricts as the parasympathetic nervous system stimulates the inner circular muscle fibres to contract.



Outer radial muscle fibres contract

DILATED PUPIL

When light is dim and the eye needs more light to see, the pupil widens as the sympathetic nervous system causes the outer radial muscle fibres to shorten.

AROUND THE EYE

When it closes, the eyelid physically protects the eye and smears lacrimal fluid, or tears, over the conjunctiva. Tears wash away dirt and dust and protect against microbes. Around the eyeball, there are six small,

strap-like muscles that attach it to the socket (orbit) in the skull bone. These extraocular or extrinsic muscles are very fast-acting and swivel, or roll, the eyeball so the eye can look up or down, inward or out.

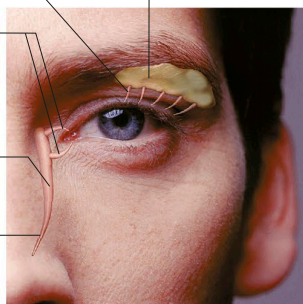
Lacrimal ducts
5–10 ducts convey fluid to the surface of the eye

Lacrimal gland
Secretes tears to keep eye clean and moist

Lacrimal canals
Collect tears draining through small holes in the corner of the eye

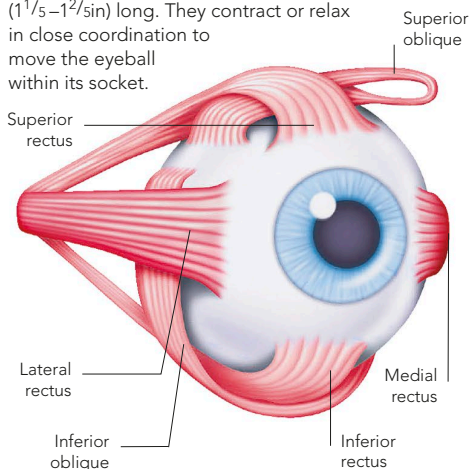
Lacrimal sac
Channels tears towards nose

Nasolacrimal duct
Opens into nasal cavity



EYE MUSCLES OF RIGHT EYE

The six extrinsic muscles are 30–35mm ($1\frac{1}{5}$ – $1\frac{2}{5}$ in) long. They contract or relax in close coordination to move the eyeball within its socket.



TEAR APPARATUS

The tear (lacrimal) gland is under the soft tissues of the outer part of the upper eyelid. It produces 1–2ml ($\frac{1}{3}$ – $\frac{2}{3}$ fl oz) of fluid daily.